

API Documentation & Submission Notes

Spring Boot · H2 (in-memory) · Swagger / OpenAPI

ENDPOINTS

Method	Endpoint	Description
POST	/api/tickets	Creates a support ticket and queues it for AI analysis.
GET	/api/tickets/{id}	Retrieves a ticket by ID, including the AI analysis once processing completes.

REQUEST / RESPONSE EXAMPLES

POST /api/tickets

REQUEST BODY

```
{
  "customerId": "CUST-301",
  "subject": "Password reset link not working",
  "description": "I clicked the password reset link in my email but it says the link has expired",
  "priority": "MEDIUM",
  "product": "Auth"
}
```

RESPONSE — 201 CREATED

```
{
  "id": 1,
  "customerId": "CUST-301",
  "subject": "Password reset link not working",
  "description": "I clicked the password reset link in my email but it says the link has expired",
  "priority": "MEDIUM",
  "product": "Auth",
  "status": "PENDING",
  "failureReason": null,
  "analysis": null,
  "createdAt": "2026-09-27T11:19:08.435Z",
  "updatedAt": "2026-09-27T11:19:08.435Z"
}
```

Analysis runs asynchronously — the ticket is created and returned immediately with status: PENDING, before the AI analysis has run.

GET /api/tickets/{id}

RESPONSE — 200 OK (after analysis completes)

```
{
  "id": 1,
  "customerId": "CUST-301",
  "subject": "Password reset link not working",
```

```
"description": "I clicked the password reset link in my email but it says the link has expired",
"priority": "MEDIUM",
"product": "Auth",
"status": "COMPLETED",
"failureReason": null,
"analysis": {
  "category": "ACCOUNT_ACCESS",
  "summary": "Customer reported an issue with subject \"Password reset link not working\" related to Auth.",
  "suggestedResponse": "Thank you for reaching out. We've received your report and our account security team is looking into it. We'll follow up as soon as we have an update.",
  "recommendedTeam": "ACCOUNT_SECURITY",
  "confidence": 0.85
},
"createdAt": "2026-09-27T11:19:08.435Z",
"updatedAt": "2026-09-27T11:19:08.829Z"
}
```

SWAGGER DOCUMENTATION:
POST:

POST /api/tickets

Parameters

No parameters

Request body `request`

`application/json`

```
{
  "customerId": "CUST-301",
  "subject": "Password reset link not working",
  "description": "I clicked the password reset link in my email but it says the link has expired",
  "priority": "MEDIUM",
  "product": "Auth"
}
```

Exclude Clear

Responses

Curl

```
curl -X POST \
  http://localhost:8080/api/tickets \
  -H 'accept: */*' \
  -H 'content-type: application/json' \
  -d '{
    "customerId": "CUST-301",
    "subject": "Password reset link not working",
    "description": "I clicked the password reset link in my email but it says the link has expired",
    "priority": "MEDIUM",
    "product": "Auth"
  }'
```

Response URL

`http://localhost:8080/api/tickets`

Server response

Code Details

201 `created`

Response body

```
{
  "id": 1,
  "customerId": "CUST-301",
  "subject": "Password reset link not working",
  "description": "I clicked the password reset link in my email but it says the link has expired",
  "priority": "MEDIUM",
  "product": "Auth",
  "status": "PENDING",
  "failureReason": null,
  "analysis": {
    "category": "ACCOUNT_ACCESS",
    "summary": "Customer reported an issue with subject \"Password reset link not working\" related to Auth.",
    "suggestedResponse": "Thank you for reaching out. We've received your report and our account security team is looking into it. We'll follow up as soon as we have an update.",
    "recommendedTeam": "ACCOUNT_SECURITY",
    "confidence": 0.85
  },
  "createdAt": "2026-09-27T11:19:08.435Z",
  "updatedAt": "2026-09-27T11:19:08.829Z"
}
```

Response headers

```
connection: keep-alive
content-type: application/json
date: Sat, 27 Sep 2025 11:20:08 GMT
keep-alive: timeout=60
transfer-encoding: chunked
```

GET:

Code Details

200

Response body

```
{
  "id": 1,
  "customerId": "CUST-301",
  "subject": "Password reset link not working",
  "description": "I clicked the password reset link in my email but it says the link has expired",
  "priority": "MEDIUM",
  "product": "Auth",
  "status": "PENDING",
  "failureReason": null,
  "analysis": {
    "category": "ACCOUNT_ACCESS",
    "summary": "Customer reported an issue with subject \"Password reset link not working\" related to Auth.",
    "suggestedResponse": "Thank you for reaching out. We've received your report and our account security team is looking into it. We'll follow up as soon as we have an update.",
    "recommendedTeam": "ACCOUNT_SECURITY",
    "confidence": 0.85
  },
  "createdAt": "2026-09-27T11:19:08.435Z",
  "updatedAt": "2026-09-27T11:19:08.829Z"
}
```

Response headers

```
connection: keep-alive
content-type: application/json
date: Sat, 27 Sep 2025 11:20:08 GMT
keep-alive: timeout=60
transfer-encoding: chunked
```

Responses

GET /api/tickets/{id}

Parameters

Name Description

id `required` Integer (int32) (path)

1

Exclude Clear

Responses

200

Response body

```
{
  "id": 1,
  "customerId": "CUST-301",
  "subject": "Password reset link not working",
  "description": "I clicked the password reset link in my email but it says the link has expired",
  "priority": "MEDIUM",
  "product": "Auth",
  "status": "PENDING",
  "failureReason": null,
  "analysis": {
    "category": "ACCOUNT_ACCESS",
    "summary": "Customer reported an issue with subject \"Password reset link not working\" related to Auth.",
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Response headers

```
connection: keep-alive
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transfer-encoding: chunked
```

PERSISTENCE VERIFICATION — H2 CONSOLE

Querying `SELECT * FROM TICKETS` against the in-memory H2 database confirms both the ticket data and the AI-generated analysis are persisted correctly end to end:

ID	Customer	Priority	Status	Category	Team	Conf.
1	CUST-301	MEDIUM	COMPLETED	Account Access	Account Security	0.85
2	CUST-303	HIGH	COMPLETED	Account Access	Account Security	0.90

DEVELOPMENT NOTES

Approximate time spent on the assessment: 7–8 hours.

AI tools used during development: roughly 25% of the development process involved AI assistance, used as a supporting productivity tool for:

- Clarifying error messages and debugging issues during development.
- Reviewing and improving test cases, and identifying additional edge cases.
- Generating sample request payloads and test data for API testing.
- Checking syntax and suggesting minor code-level improvements.
- Troubleshooting configuration and environment-related issues.

ASSUMPTIONS & LIMITATIONS

- Each request creates exactly one ticket — there is no bulk or batch ticket submission.
- One LLM call produces one complete analysis; there is no support for partial, streaming, or multi-turn analysis.
- Persistence: H2 runs in-memory (`ddl-auto: create-drop`), so all ticket data is lost on every restart. This was a deliberate choice so the project runs with zero external setup, not an oversight.
- No authentication or authorization — both endpoints are fully open. Anyone with network access to the service can create or read any ticket.
- No retry or backoff on transient LLM failures — a timeout or transient provider error currently fails the ticket immediately rather than retrying with backoff.
- Next step: add API versioning and a stable, documented contract (once OpenAPI generation is fixed) before external consumers depend on this service.